DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2024-2025: PERIODIC ASSESSMENT-III

Subject: Science

SET–I

TIME: $1\frac{1}{2}$ Hrs. **M.M.: 40**

Date: 11/12/24 **Name:** Mohd. Hifzaan Khan **Class/Sec IX** D

General Instructions:

Read the following instructions carefully and follow them:

- **The question paper comprises of three sections, A, B and C. You have to attempt all the sections.**
- **All questions are compulsory.**
- **Q1 to Q10 carry one mark each.**
- **Q11 to Q13 are very short type questions and carry two marks each.**
- **Q14 to Q16 are short type questions and carry three marks each.**
- **Q17 to Q19 are long type questions and carry five marks each.**

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SECTION A

Select the most appropriate answer. **(1x6=6)**

1. Which of the following species is used for pearl cultivation?
(a) Carps
(b) Oysters
(c) Prawns
(d) Bhetki

2. While comparing the pressure exerted by the three different faces of a metal cuboid of dimensions 10 x 20 x 30 cm, which observation is correct?
(a) Pressure exerted by all faces is same
(b) Pressure exerted by the face 10 x 20 is maximum
(c) Pressure exerted by the face 10 x 30 is maximum
(d) Pressure exerted by the face 30 x 20 is maximum

3. Particles which have more electrons than the normal atoms-
(a) Molecules
(b) Ions
(c) Cation
(d) Anion

4. Nitrogen, phosphorus, calcium are examples of:
(a) Micronutrients
(b) Green manure
(c) Macronutrients
(d) Both (a) and (b)

P > 2, 5, 8, 12, 16, 21
B > 1, 7, 9, 11, 14, 20
C > 3, 4, 6, 10, 13, 15, 22

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5. The work done on an object does not depend upon the
(a) Displacement
(b) Force applied
(c) Angle between force and displacement
(d) Initial velocity of the object

6. Atomic radius is measured in-
a. Centimeters
b. Meters
c. Nanometers
d. Micrometers

Direction (Q7 to 10): In the following questions, there are two statements marked as Assertion and Reason. Mark your answer as per the codes provided below: **(1x4=4)**

(a)Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
(b)Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(c) Assertion is true but Reason is false.
(d) Assertion is false but Reason is true.

7. **Assertion:** Berseem, oats, sorghum etc. are called fodder crops.
Reason: These crops are raised as food for the livestock. a

8. **Assertion:** If velocity of a body is doubled, its kinetic energy becomes four times.
Reason: Kinetic energy is proportional to square of velocity. a b

9. **Assertion:** Cattle husbandry is done for two purposes-milk and draught labour.
Reason: Milk production depends on lactation period. b

10. **Assertion:** Atomic mass of Sulphur is 32u.
Reason: An atom of Sulphur is 32 times heavier than one-twelfth of the mass of carbon – 12 Atom. a

SECTION B

11. Differentiate between the following: (give one difference) (2)
a. Broilers and layers
b. Intercropping and Mixed cropping

12. A body of volume 3 cm^3 is completely immersed in water. What is the buoyant force acting on the body. (2)

13. (a)In Carbon dioxide, carbon and oxygen are always present in the ratio 3:8 by mass. State the Law which explains the above statement. (2)
(b) During the formation of Ammonia, what mass of hydrogen gas would be required to react completely with 28 g of Nitrogen gas

SECTION C

14. Define the following terms: (3)
a. Genetic manipulation b. Macronutrients c. Composite fish culture

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15. Consider the following chemical reaction:

$$CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + CO_2 + H_2O$$

8.4g of $NaHCO_3$ was treated with 6.0g of CH_3COOH. The residue was found to weigh 10.0g.

(a) Determine the mass of carbon dioxide escaped in the reaction. (1)

(b) Write the chemical name of the following compounds (2)

(i) KOH (ii) SO_2

16. i) State the Archimedes principle and give two applications of this principle. (2)

ii) Define Relative Density. (1)

SECTION D

20. Read the following text carefully and answer the questions that follow:

A farmer has grown wheat on his field consecutively two times but when the third time he grows wheat on the same field the quality of the wheat was not up to the desired level. To improve the quality of his crops he uses chemical fertilizer but the condition of the crop became worse. One of his friends told him to grow a different variety of crops after wheat so as to grow two or three crops in a year with good harvests and use biological manure in place of chemical fertilizers which he prepares by animal excreta and plant waste to get the good quality of crops. He also told him about the good storage of his grains to protect them from the biotic and abiotic losses as in agriculture storage losses are very high.

a. What is crop rotation? (1.5)

b. What is the advantage of using biological manure over chemical fertilizers? (1.5)

c. Enlist the biotic and abiotic losses. (2)

OR

Is it possible for the farmer to grow two crops at the same time if yes what is the requirement? (2)

21. (a). A block of mass 2kg slides down an inclined plane of inclination 30°. Find the work done by the force of gravity as the block slides through 1m. [Take g = 10 ms^{-2}]. (2)

(b). Write the relation between Kinetic Energy and Momentum. (1)

(c). How is kWh related to Joule. (1)

(d). Define power. What is its SI unit ? (1)

22. i. Write the chemical formula of following compound using criss-cross method: (2)

(a) Sodium nitrate (b) Calcium carbonate

ii. Calculate the molecular mass of the following compounds (2)

(a) MgO (b) $CaCl_2$

iii. Compute the ratio by mass of combining elements in Calcium oxide. (1)